

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**



FILED

08/15/22

03:47 PM

R2207005

Order Instituting Rulemaking to
Advance Demand Flexibility
Through Electric Rates.

Rulemaking 22-07-005
(Filed July 14, 2022)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ORDER
INSTITUTING RULEMAKING TO ADVANCE DEMAND FLEXIBILITY THROUGH
ELECTRIC RATES**

Edward Burgess
Senior Policy Director

Zach Woogen
Policy Manager

VEHICLE-GRID INTEGRATION COUNCIL
10265 Rockingham Dr.
Suite #100-4061
Sacramento, California 95827
Telephone: (510) 665-7811
Email: vgicregulatory@vgicouncil.org

August 15, 2022

**BEFORE THE PUBLIC UTILITIES COMMISSION
OF THE STATE OF CALIFORNIA**

Order Instituting Rulemaking to
Advance Demand Flexibility
Through Electric Rates.

Rulemaking 22-07-005
(Filed July 14, 2022)

**COMMENTS OF THE VEHICLE-GRID INTEGRATION COUNCIL ON THE ORDER
INSTITUTING RULEMAKING TO ADVANCE DEMAND FLEXIBILITY THROUGH
ELECTRIC RATES**

In accordance with the Rules of Practice and Procedure of the California Public Utilities Commission (“Commission”), the Vehicle-Grid Integration Council (“VGIC”) ¹ hereby submits these comments on the *Order Instituting Rulemaking to Advance Demand Flexibility Through Electric Rates* (“OIR”), adopted by the Joint Commissioners on July 14, 2022.

I. INTRODUCTION.

VGIC commends the Commission for issuing the OIR and appreciates the attention given to advancing demand flexibility, especially considering the growing integration of distributed energy resources (“DERs”), including electric vehicles (“EVs”), into the electric system. VGIC is excited to see the continuation of a conversation that began last year surrounding the universally accessible, dynamic, and economic signal (“UNIDE”) framework presented by Commission Staff at a workshop on May 25, 2021. Shaping DER dispatch and load generally through retail rates

¹ VGIC member companies and supporters include American Honda Motor Co., Inc., dcbel, Enel X North America, Inc., ENGIE NA, Fermata Energy, FlexCharging, FLO EV Charging, Ford Motor Company, FreeWire Technologies, Inc., General Motors Company, Kaluza, Nissan Group of North America, Nuvve Holding Corporation, Sacramento Municipal Utility District, Stellantis N.V., Sunrun, Switch EV Ltd, The Mobility House, Toyota Motor North America, Inc., Veloce Energy, Inc., Wallbox USA Inc., and WeaveGrid. The views expressed in these Comments are those of VGIC, and do not necessarily reflect the views of all individual VGIC member companies or supporters. (<https://www.vgicouncil.org/>)

provides a path to incentivize these resources to beneficially serve the grid while also allowing for costs to be recovered if load is shaped uneconomically. While other mechanisms, such as wholesale market integration, can also help achieve these goals, barriers have been faced when developing these other pathways. On the other hand, dynamic pricing rates have already been deployed in California. They will soon be expanded, for example, through Pacific Gas & Electric's Day-Ahead Hourly Real-Time Pricing ("DAHRTTP") commercial EV rate.

In support of this OIR, Commission Staff released a whitepaper, "Advanced Strategies for Demand Flexibility Management and Customer DER Compensation," outlining a California Flexible Unified Signal for Energy ("CalFUSE") framework for future dynamic rates in the state incorporating real-time pricing ("RTP") for generation and distribution capacity, as well as bidirectional electricity prices and hedging subscription and transactive products.

EVs are well-positioned to take advantage of dynamic pricing and load flexibility. Vehicle Grid Integration ("VGI") strategies, including managed charging and discharging, can unlock EVs as a strategic grid asset with the right price signals. These price signals can directly inform customer behavior or integrate with automated charge management systems for a "behind the scenes" customer experience.

VGIC agrees with the Commission that "[t]he rapid growth of electric end uses – including electric vehicle charging, distributed energy resources, and building decarbonization – presents new challenges and opportunities for coordinating demand flexibility to meet system needs on a regular basis."² To this end, VGIC offers these comments and recommendations to ensure holistic

² OIR at 1-2.

consideration of California’s climate goals and continue near-term progress on enabling dynamic rates and managed charging programs for EV customers.

II. BACKGROUND & INTEREST IN PROCEEDING.

VGIC is a 501(c)6 membership-based advocacy group committed to advancing the role of EVs and VGI through policy development, education, outreach, and research. VGIC supports the transition to a decarbonized transportation and electric sector by ensuring the value from EV deployments and flexible EV charging and discharging is recognized and compensated to achieve a more reliable, affordable, and efficient electric grid. VGIC currently has 22 members, including automotive original equipment manufacturers (“OEMs”) and EV service providers (“EVSPs”), and has a direct interest in shaping the policies, procedures, and rules needed to modernize the electric grid for a high DER future. VGIC has been an active participant in related rulemakings such as the Transportation Electrification or “DRIVE” (R.18-12-006), Self-Generation Incentive Program (R.20-05-012), Microgrids and Resiliency (R.19-09-009), Streamlining Interconnection of DERs (R.17-07-007), Emergency Summer Reliability (R.20-11-003), and High DER Future (R.21-06-017) proceedings.

III. COMMENTS ON THE PROPOSED SCOPE.

In recognition of the anticipated growth of DERs, including VGI-capable EVs, VGIC appreciates the magnitude of the challenge facing California’s electric grid. Previously, load was seen as relatively inflexible demand that had to be met by supply; however, as multiple end-uses become electrified, including transportation, shaping load in ways that are useful to the grid becomes both more feasible and more critical to maintaining electric reliability and affordability. To harness and shape this load, VGIC supports the Commission opening this proceeding to consider how to advance demand flexibility.

A. Rate design principles should consider non-electric sector state goals, including transportation sector emissions reductions.

VGIC generally supports the ambitious nature of this proceeding and believes that considering rate design principles at this juncture is appropriate. The basic principles of maintaining a reliable and affordable grid should remain. However, the Commission should consider updating certain principles that do not meet the needs of today's evolving grid, including the following principle: "Rates should encourage the reduction of coincident and non-coincident peak demand."³ Specifically, increased non-coincident peak demand, for example, from daytime EV charging load, does not necessarily increase costs or pose reliability risks. Therefore, rates need not encourage reductions to non-coincident peak demand. As low-cost renewable energy supply grows, increasing the non-coincident peak demand of an individual customer through daytime EV charging can benefit the grid by absorbing excess solar production, including self-consumption at a site with rooftop solar photovoltaic generation.

To this end, VGIC urges the Commission to consider how rates can advance other goals in the state of California, such as greenhouse gas ("GHG"), climate, and air pollution goals that extend beyond the electric sector. The transportation sector is California's largest source of GHG emissions, accounting for almost 40% of statewide emissions in 2019.⁴ To address this, Governor Newsom's Executive Order N79-20 targets 100% zero-emission vehicle

³ D.12-06-029 at 12.

⁴ California Air Resources Board. *California Greenhouse Gas Emissions for 2000 to 2019: Trends of Emissions and Other Indicators*. Available at: https://ww3.arb.ca.gov/cc/inventory/pubs/reports/2000_2019/ghg_inventory_trends_00-19.pdf

(“ZEV”) market share for in-state sales of new passenger cars and trucks by 2035 and 100% ZEV medium- and heavy-duty vehicles by 2045.⁵

When considering California’s climate goals, it is appropriate for the Commission to take a holistic view of emissions across the state and consider how rate design can promote transportation electrification (“TE”) to yield lower transportation sector GHG emissions. The Commission should therefore consider adding a rate design principle to directly encourage TE, which supports affordability, reliability, and emissions reductions for both the electric and transportation sectors.

B. If the Commission chooses to pursue a unified technology-agnostic rate, progress on EV-specific rate offerings should not be delayed or prematurely ended.

California load-serving entities (“LSE”) have been incorporating aspects of the proposed CalFUSE framework into EV-specific rates over the past several years. All three investor-owned utilities (“IOUs”) offer residential and commercial EV-specific rates to encourage beneficial EV charging behavior and recover appropriate costs of service. Generally, these rates are high-differential time-of-use (“TOU”) rates that could be applied to the entire customer account or a separately metered EV. The Commission’s recent Submetering Decision (“D.”) 22-08-024 provides an alternative to separately metered and whole-site TOU rates by utilizing the EVSE to measure charging load. EV-specific TOU rates, whether separately metered, submetered, or whole-site, promote demand flexibility and introduce customers to the concept of time-differentiated rates.

⁵ Executive Order N-79-20. Available at: <https://www.gov.ca.gov/wp-content/uploads/2020/09/9.23.20-EO-N-79-20-Climate.pdf>

In addition to the EV-specific TOU rates currently offered by each IOU, San Diego Gas and Electric (“SDG&E”) is piloting an innovative VGI Rate option as part of its Power Your Drive program. This rate includes a California Independent System Operator (“CAISO”) wholesale market day-ahead energy price component with a day-of adjustment, a peak demand adder, and a distribution circuit demand adder. The April 2021 Power Your Drive Research Report shows that the VGI rate successfully aligns charging load with grid needs, “reducing load when CAISO prices rise or system or circuit adders are applied.”⁶ Additionally, Pacific Gas and Electric (“PG&E”) plans to offer a dynamic day-ahead, hourly, real time price (“DAH RTP”) rate option to all eligible commercial EV customers with a target launch date of October 2023.⁷ Notably, this is not a capped or experimental rate pilot like SDG&E’s VGI Rate or TeMix RATES Pilot mentioned in the CalFUSE whitepaper, but rather a full-scale optional rate open to all eligible commercial customers. In June 2022, several parties in PG&E’s DAH RTP application jointly filed a settlement agreement proposing an export compensation mechanism for non-net energy metering (“NEM”) DAH RTP customers, which could also be offered to customers as early as October 2023.⁸ The export compensation mechanism, which provides a bidirectional price signal for marginal energy and marginal generation capacity costs, is pending before the Commission. Lastly, Resolution E-5192 directs PG&E to file a Tier 2 Advice Letter by September 2, 2022, proposing a dynamic rate

⁶ *Power Your Drive Research Report*. SDG&E. April 2021. Pg 19.

<https://www.sdge.com/sites/default/files/regulatory/SDG%26E%20FINAL%20Power%20Your%20Drive%20Research%20Report%20April%202021.pdf>

⁷ *Decision Authorizing PG&E to Implement an Optional Day-Ahead Real Time Rate for Commercial EV Customers*. November 19, 2021.

<https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M424/K557/424557371.PDF>

⁸ *Joint Motion of the Public Advocates Office, VGIC, Electrify America, and PG&E for Adoption of Joint Settlement Agreement in PG&E’s DAH RTP Commercial EV Proceeding on Non-NEM Export Compensation Pilot*. June 17, 2022.

<https://docs.cpuc.ca.gov/PublishedDocs/Efile/G000/M488/K538/488538216.PDF>

for its approved VGI pilots. The rate design requirements that PG&E must meet, detailed in Section 10 of the Resolution, closely follow the CalFUSE framework, including bidirectional pricing, dynamic marginal energy prices based on wholesale electricity costs, and dynamic marginal capacity prices based on grid utilization. These developments, specifically PG&E's DAHRTP rate option, represent critical strategies to promote flexible EV charging demand and by no means should be delayed, altered, or prematurely concluded.

The CalFUSE framework contemplates a unified, technology-agnostic rate to simplify the menu of options and reduce complexity for customers. VGIC appreciates the underlying intent of this approach and understands how it may unlock widespread demand flexibility in California. However, if the Commission does decide to adopt a “one size fits all” solution, VGIC strongly recommends the Commission use a phased approach that does not abruptly and prematurely halt critical progress on the above-mentioned EV-specific rate approaches. At this early stage of the market, technology-specific rates, including SDG&E's PYD VGI rate, PG&E's planned DAHRTP rate, and PG&E's upcoming dynamic rate design proposal according to Resolution E-5192, remain appropriate and should be offered in the interim period until - and perhaps even alongside – the phase-in of a potential universal CalFUSE rate.

C. The Commission should consider looking beyond a “one size fits all” solution and instead consider a range of rate design and programmatic approaches.

The 2019-2020 VGI Working Group identified 320 unique VGI use cases, as different EV customers and charging site hosts have different mobility needs, willingness and ability to manage charging, and technological capabilities.⁹ A single solution may not be a good fit for

⁹ *Final Report of the California Joint Agencies Vehicle-Grid Integration Working Group*. June 30, 2020. <https://gridworks.org/wp-content/uploads/2020/07/VGI-Working-Group-Final-Report-6.30.20.pdf>

all passenger EV drivers, fleets (e.g., electric school buses, last-mile delivery, etc.), and public fast charging sites. Therefore, a flexible portfolio of rate design options and programmatic managed charging approaches should be implemented to accommodate the vast array of flexible EV charging use cases.

With this in mind, VGIC strongly recommends this proceeding not detract from Commission efforts to progress programmatic EV demand flexibility strategies. While this proceeding is focused on advancing demand flexibility through electric rates, VGIC believes it is critical to continue offering non-rate options, including the Emergency Load Reduction Program EV/VGI Aggregation Pathway, PG&E's Low Carbon Fuel Standard-funded Resilient Charging Pilot, SCE's proposed Large-Scale Residential V1G Pilot, and SDG&E's proposed EV DR Pilot. In complement to EV-specific rates, these programmatic approaches represent significant drivers for VGI market development and facilitate participation from the broadest possible set of customers.

IV. CATEGORIZATION, HEARINGS, AND SCHEDULE.

VGIC supports the categorization and proposed schedule of this proceeding and agrees with the preliminary determination finding the need for hearings.

V. NOTICES.

Services of all notices and communications in this proceeding should be directed to the following VGIC representative:

Ed Burgess
Senior Policy Director
VEHICLE-GRID INTEGRATION COUNCIL
10265 Rockingham Dr.
Suite #100-4061
Sacramento, CA 95827

Telephone: (510) 665-7811
Email: vgicregulatory@vgicouncil.org

VI. CONCLUSION.

VGIC appreciates the opportunity to submit these comments on the OIR. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

A handwritten signature in black ink, appearing to read "Edward Burgess", with a stylized, cursive script.

Edward Burgess
Senior Policy Director
VEHICLE-GRID INTEGRATION COUNCIL

Date: August 15, 2022